

PROJECT PHASE 2: EXPLORATORY DATA ANALYSIS & FEATURE ENGINEERING

Deadline: Week 7

Weight: 15% of the project grade

Penalty: Each delay is penalized with 3 points

Maximum allowed delays: 2

Objective

The goal of this phase is to **prepare the dataset for intelligent processing** by applying:

- Exploratory Data Analysis (EDA)
- Data cleaning
- Feature engineering techniques

Students must go beyond basic exploration and identify meaningful patterns, improve data quality, and construct relevant features for modelling.

Coding (EDA, Data Processing & Feature Engineering)

In this phase, you must **extend your work from Phase 1** and transform the dataset into a **clean, structured, and model-ready format**.

◆ **1. Data Cleaning**

Minimum requirements:

- Detect and treat:
 - outliers
 - duplicates
 - inconsistent values
- Fix data type issues (e.g. strings → datetime, categorical encoding)

You must **justify your choices in code comments**

◆ **2. Feature Selection**

- Identify relevant features for the problem
- Remove:
 - irrelevant features

- redundant features (high correlation / multicollinearity)

Provide **clear reasoning for keeping/removing features**

◆ 3. Feature Extraction

- Extract useful information from raw data
- Apply the appropriate techniques: e.g., PCA, t-SNE, etc

Examples:

- text → length, word count, sentiment score
- categorical → frequency encoding

◆ 4. Feature Transformation / Construction

- Create new features that improve representation

Examples:

- normalization / standardization
- log transformations
- ratios / aggregations
- interaction features

At least **2–3 meaningful engineered features are required**

◆ 5. Univariate Analysis

- Analyze individual variables:

Examples:

- distributions (histograms, KDE)
- boxplots
- skewness

Identify:

- outliers
- skewed distributions
- anomalies

◆ 6. Bivariate Analysis

- Analyze relationships between variables

Examples:

- correlation matrix
- scatter plots
- grouped statistics
- categorical vs numerical analysis

Identify:

- strong relationships
- potential predictors
- hidden patterns

◆ 7. Problem-Relevant EDA

- At least **3–5 visualizations directly linked to the project goal**

Each visualization must:

- be clearly labeled
- include titles and axes
- include **1–2 sentences interpretation**

Report

Please continue to work on the report template provided:

<https://github.com/SergiuLimboi/Intelligent-techniques-for-processing-structured-and-large-data/tree/main/Report%20template>

The report must include:

1. Exploratory Data Analysis (EDA)

- Summary of dataset exploration
- Key distributions and relationships
- Important observations from univariate and bivariate analysis

2. Data Quality

- Missing values (patterns + handling strategy)

- Outliers (detection + treatment)
- Data inconsistencies

👉 Must include:

- explanation of issues
- justification of decisions

3. Important Patterns

- Key findings discovered during analysis

Examples:

- correlations
- clusters / segments
- trends over time

👉 This is a **critical section**:

students must extract **insights, not just show plots**

4. Feature Engineering

- Description of:
 - selected features
 - created features
 - transformations applied

👉 Explain:

- why each feature matters
- expected impact on modelling
-

📦 Deliverables (Mandatory)

Report:

- LaTeX source file
- PDF version

Code:

- Jupyter Notebook or Python scripts
- Must be runnable
- Must include comments and explanations

Repository:

- All work must be committed to the corresponding GitHub Classroom team repository

Submission rule:

For each deliverable upload to GitHub:

- PDF with report
- PDF with code + outputs (export notebook to PDF)

👉 Must be uploaded **before the laboratory in Week 7**

Grading

- Report: **50%**
- Implementation (code): **50%**

Final grade = weighted average

Grades will be provided after evaluation by course & lab instructors

Important

- Work will NOT be evaluated if not committed to GitHub Classroom
- Ensure all required files are present before deadline

Grading Rubric — Code (EDA & Feature Engineering)

Section	Points	Full points	Partial points	No points
1 . Data cleaning	1.5	Missing values, outliers, duplicates handled correctly with justification	Partial handling or no justification	No cleaning
2 . Feature selection	1.5	Relevant features selected with reasoning (correlation, domain knowledge) Use of methods: filter methods, wrapper, embedded	Selection done but no justification	No selection
3 . Feature extraction	1.5	Meaningful features extracted from raw data .	Basic extraction, limited usefulness.	No extraction
4 . Feature transformation	1.5	Proper transformations applied (scaling, encoding, new features)	Transformations applied but unclear impact	No transformation
5 . Univariate analysis	1.5	Distributions analyzed and interpreted	Plots without interpretation	No analysis
6 . Bivariate analysis	1.5	Relationships identified and explained	Basic plots, no insights	No analysis
7 . Problem-relevant visualisations	1	3–5 relevant, labeled, interpreted charts .	Generic or unlabeled charts.	<3 or irrelevant
Total	10 points			

Grading Rubric — Report Phase 2

Section	Points	Full points	Partial points	No points
1 . EDA description	2	Clear, structured, insightful analysis (with relevant plots)	Basic description	Missing
2 . Data quality	1	Issues clearly identified + justified handling	Issues mentioned but not analyzed	Missing
3 . Important patterns	3	Strong insights with interpretation	Observations but shallow	No insights
4 . Feature engineering	3	Well-justified features and transformations with proper examples	Features listed but not explained	Missing
5 . Clarity & structure	1	Well written, structured	Plots without interpretation	Poor
Total	10 points			